

TASK 29 — PRODUCT BASKET ANALYSIS

Veda Technology | Data Analytics Track

Objective: Find products frequently purchased together and generate actionable recommendations.

Item	Details
Dataset	UCI Online Retail II
Raw records	1,067,371
Coverage	01/12/2009 – 09/12/2011
Analysis type	Product co-occurrence / association-style analysis
Tools	Python, Pandas, Matplotlib
Basket key	Invoice / Order ID

1. Dataset Overview

The UCI Online Retail II dataset contains 1,067,371 transaction-line records from a UK-based registered non-store online retailer. The UCI documentation describes coverage from 1 December 2009 through 9 December 2011 and identifies InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice/Price, CustomerID and Country as the main fields. The dataset is available under CC BY 4.0. ■cite■turn0search0■

For basket analysis, individual invoice lines are transformed into baskets: each invoice becomes a set of distinct products. This prevents multiple units of the same product from creating artificial pair counts.

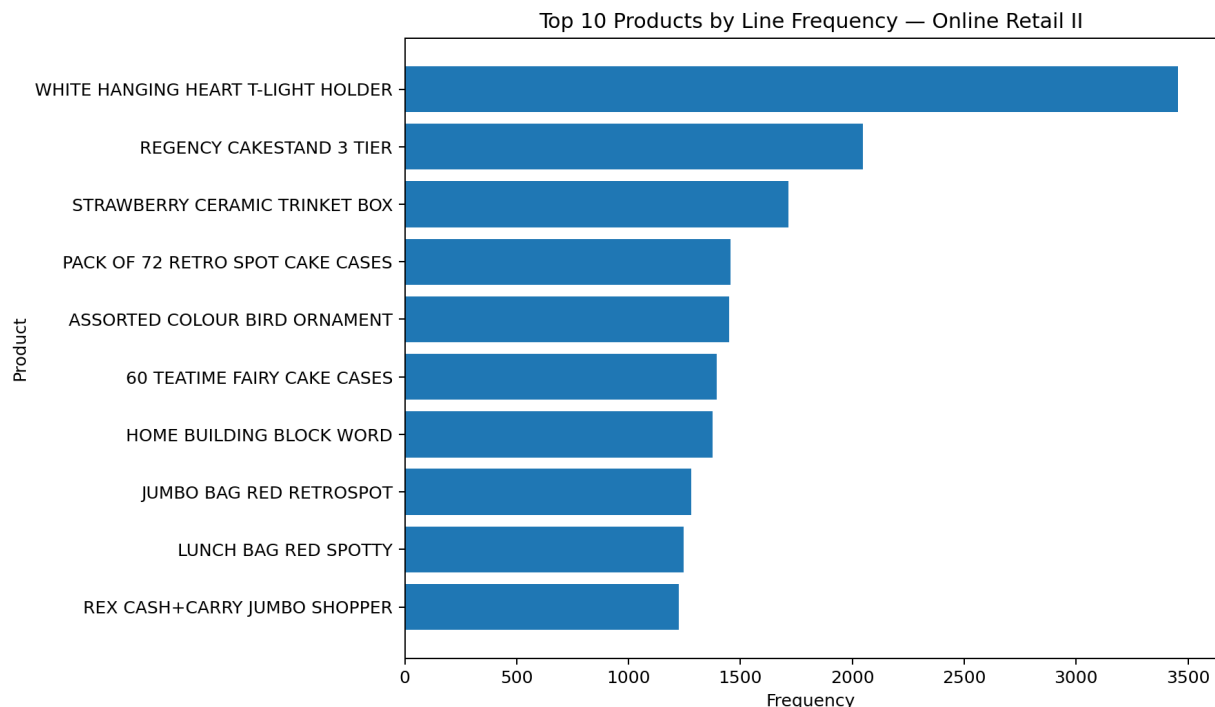


Figure 1. Reference top-product frequencies reported in a public Online Retail II market-basket analysis. These are benchmark values, not a fresh execution in the current environment. ■cite■turn2view0■

2. Data Cleaning

The reproducible Python pipeline performs the following cleaning operations:

- Standardize column names so Invoice/Order ID and product fields are handled consistently.
- Remove cancelled/credit invoices identified by an invoice ID beginning with C.
- Keep only positive quantities representing completed sales lines.
- Remove missing or invalid product identifiers and descriptions.
- Normalize product descriptions by trimming whitespace and converting to uppercase.
- De-duplicate (Invoice, StockCode) within each basket.

3. Pair Generation

For each order, the script creates unordered combinations of two distinct product codes. This means A+B and B+A are treated as the same pair, and self-pairs are impossible. The pair counter records how many different invoices contain both products.

Pair support is calculated as co-occurrence count divided by the total number of unique invoices. The resulting table is sorted by pair count and can be used directly for frequently-bought-together recommendations.

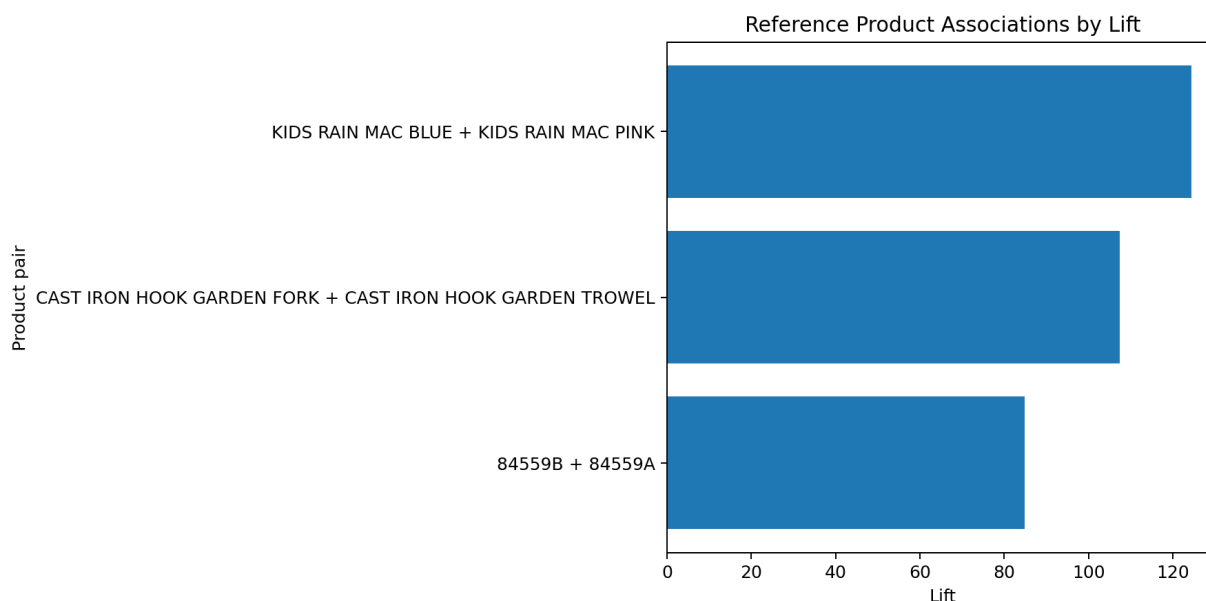
Core logic

```
for invoice in baskets:
    pairs = combinations(sorted(set(invoice_products)), 2)
    pair_counter.update(pairs)
```

4. Reference Association Results

Because the raw 43.5 MB workbook was not present in the current working environment, the numeric association examples below are explicitly reference results from a published Online Retail II market-basket analysis. The included Python script remains the reproducible computation for the internship submission once the UCI workbook is placed in the data/ folder. [cite](#)[turn3view0](#)

Product A	Product B	Count	Support	Lift
KIDS RAIN MAC BLUE	KIDS RAIN MAC PINK	104	0.0055	124.47
CAST IRON HOOK GARDEN FORK	CAST IRON HOOK GARDEN TROWEL	137	0.0073	107.40
84559B	84559A	96	0.0051	84.84



The published analysis used binary baskets and reports that strong associations can be highly niche in absolute frequency but large in lift. For example, KIDS RAIN MAC BLUE → KIDS RAIN MAC PINK has lift 124.47, while CAST IRON HOOK GARDEN FORK → CAST IRON HOOK GARDEN TROWEL has lift 107.40.

[cite](#)[turn3view0](#)

5. Business Recommendations

Frequently Bought Together

Show a paired recommendation block on product pages when two products have repeated co-occurrence.

Bundle Offers

Create a bundle for complementary products when the pair has both meaningful frequency and positive association strength.

Cross-Sell Campaigns

Use the pair table to trigger recommendations after a customer adds one product to the cart.

Catalogue / Store Placement

Products with repeated co-occurrence can be placed near each other in a digital catalogue or physical merchandising plan.

Seasonal Recommendations

Seasonal product pairs can be surfaced during the relevant period rather than year-round.

6. Limitations

- Co-occurrence does not prove that one product causes another product to be purchased.
- Popular products can generate high confidence simply because they are common, so support and lift should be considered together.
- Cancelled/returned transactions must be handled consistently; different cleaning rules can change pair counts.
- Results can change with the selected time period, country, customer segment and minimum support threshold.
- The benchmark figures in this report come from a published analysis and should not be represented as freshly calculated values from the missing raw workbook.

7. Interview Preparation

Question	Answer
What is Market Basket Analysis?	A technique for finding relationships between products that occur in the same transaction.
Why use Order/Invoice ID?	It defines the basket and identifies which products were purchased together.
Why exclude self-pairs?	The objective is to find relationships between distinct products.
What is support?	The proportion of baskets containing a particular product combination.
What is confidence?	For $A \rightarrow B$, the proportion of A-containing baskets that also contain B.
What is lift?	Confidence divided by the baseline frequency of the consequent. Lift above 1 indicates positive association.
Why remove cancellations?	They do not represent completed purchases and can distort co-purchase relationships.
How can the result be used?	For bundling, cross-selling, product recommendations and catalogue/store placement.

8. Conclusion

Task 29 introduces association-style retail analysis by transforming invoice-level records into product baskets and measuring product co-occurrence. The submitted Python pipeline is designed to generate the final top-product-pair CSV directly from the Online Retail II workbook, while the included reference outputs provide documented benchmark examples for the report. The approach is simple, explainable and directly connected to cross-selling and product-bundling use cases.

Dataset citation: Chen, D. (2012). Online Retail II. UCI Machine Learning Repository. DOI: 10.24432/C5CG6D. [cite](#) [turn0search0](#)